the direction of magnetic field lines is shown in Figure 3.37. Since the wire is geometrically cylindrical in shape and symmetrical about its axis, we construct an Ampèrian loop in the form of a circular shape at a distance $r$ from the centre of the conductor as shown in Figure 3.37. From the Ampère's law, we get

$$\oint_C \vec{B}.d\vec{l} = \mu_\circ I$$

where $d\vec{l}$ is the line element along the Amperian loop (tangent to the circular loop). Hence, the angle between magnetic field vector and line element is zero. Therefore,

$$\oint_C B\,dl = \mu_\circ I$$

where $I$ is the current enclosed by the Ampèrian loop. Due to the symmetry, the magnitude of the magnetic field is uniform over the Ampèrian loop. Hence

$$B\oint_C dl = \mu_\circ I$$

For a circular loop, the circumference is 2πr, which implies,

$$B\int_0^{2\pi r} dl = \mu_\circ I$$

$$B\cdot 2\pi r = \mu_\circ I$$

$$B = \frac{\mu_\circ I}{2\pi r}$$

In vector form, the magnetic field is

$$\vec{B} = \frac{\mu_\circ I}{2\pi r}\hat{n}$$

where $\hat{n}$ is the unit vector along the tangent to the Ampèrian loop as shown in the Figure 3.37.

## EXAMPLE 3.15

Compute the magnitude of the magnetic field of a long, straight wire carrying a current of 1 A at distance of 1m from it. Compare it with Earth's magnetic field.

### *Solution*

Given that $I$ = 1 A and radius $r$ = 1 m

$$B_{straightwire} = \frac{\mu_\circ I}{2\pi r} = \frac{4\pi\times10^{-7}\times1}{2\pi\times1} = 2\times10^{-7}\ \text{T}$$

But the Earth's magnetic field is $B_{Earth} \sim 10^{-5}$ T

So, $B_{straightwire}$ is one hundred times smaller than $B_{Earth}$.

**Solenoid**

A solenoid is a long coil of wire closely wound in the form of helix as shown in Figure 3.38. When electric current is passed through the solenoid, the magnetic field is produced. The magnetic field of the solenoid is due to the superposition of magnetic fields of each turn of the solenoid. The direction of magnetic field due to solenoid is given by right hand palm-rule.

Inside the solenoid, the magnetic field is nearly uniform and parallel to its axis whereas, outside the solenoid the field is negligibly small. Based on the direction of the current, one end of the solenoid behaves like North Pole and the other end behaves like South Pole.

The current carrying solenoid is held in right hand. If the fingers curl in the direction of current, then extended thumb gives the direction of magnetic field of current carrying solenoid. It is shown in



UNIT-3(XII-Physics_Vol-1).indd 163 14-02-2020 20:04:01